

MIGRATING FROM QNX TO WIND RIVER

WIND RIVER WEBINAR

JULY 2024

WINDRIVER



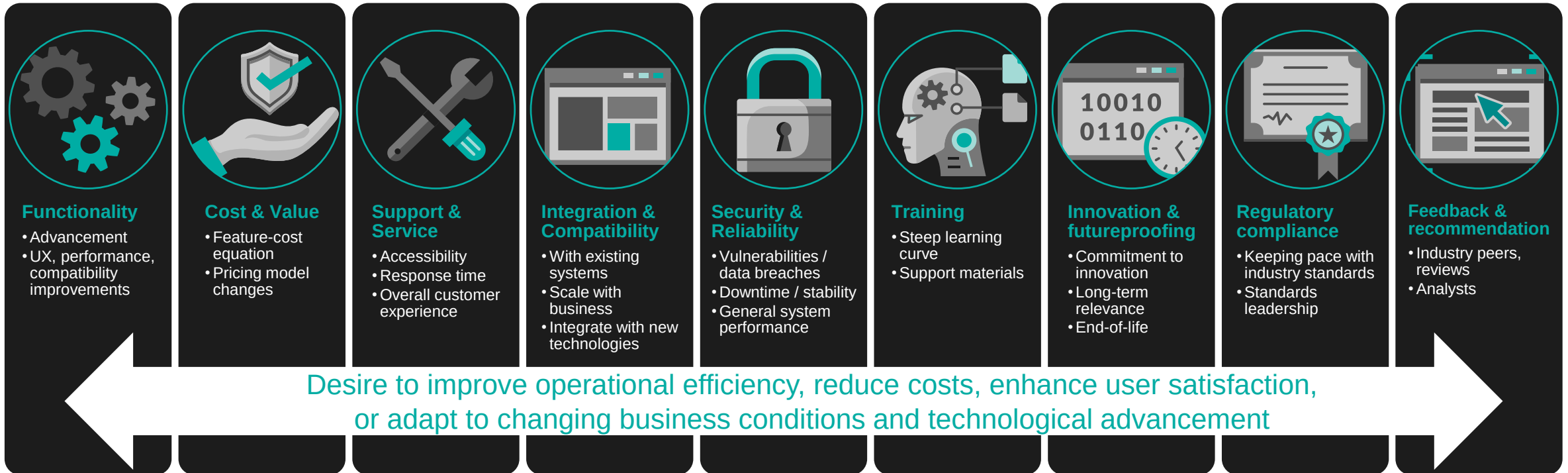
Agenda

The case for migration

Considerations

Beyond Migration

WHAT FUELS A MIGRATION DECISION?



VXWORKS CAPABILITIES AND ADVANTAGES

The edge of performance

- Native TSN
- Efficient AI/ML integration

Scales with demand

- Multicore support, wide range of BSPs
- Mixed criticality - Helix Virtualization Platform

Proven functional safety

- Cert evidence including ISO 26262 ASIL-D

Superior by design

- Proven kernel design, WR SDL, WR Security Center

Redefining time-to-market

- AWS Graviton
- OCI Containers
- WR Studio Developer



HIGH PERFORMANCE, SCALABILITY, AND DETERMINISM



PROVEN SAFETY, SECURITY, AND RELIABILITY



REDUCED COST AND TIME-TO-MARKET

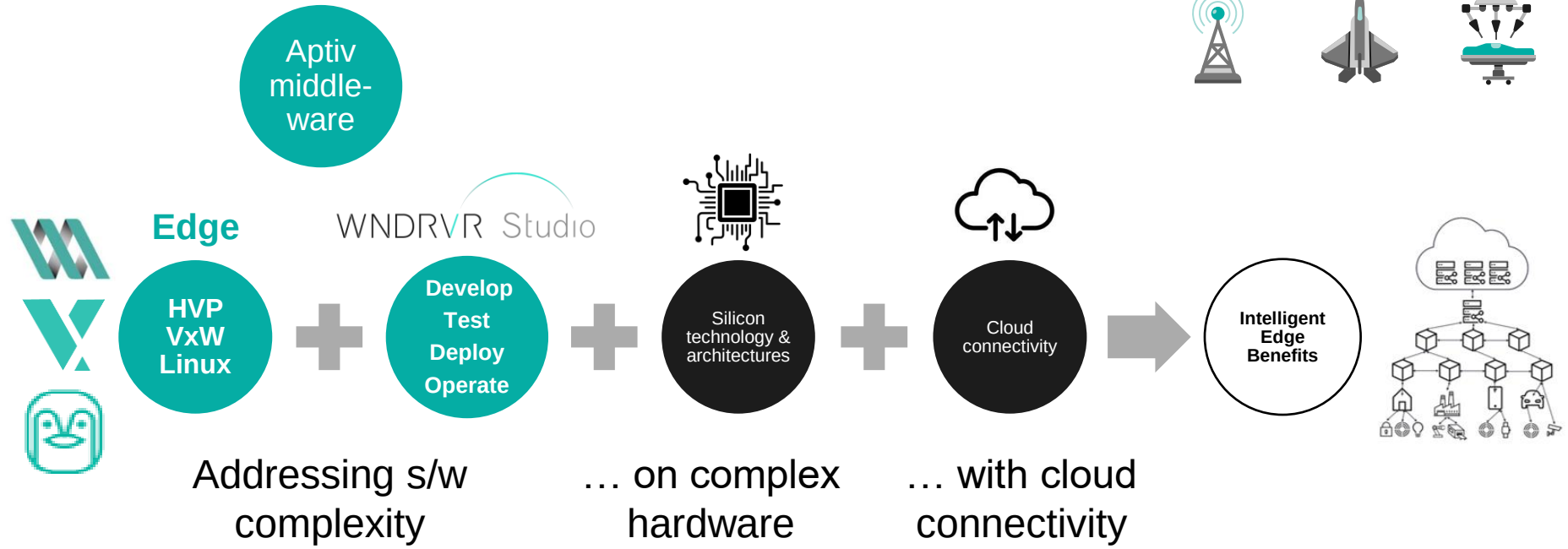
VxWorks® enables the cost-effective development of the most advanced real-time systems, significantly reducing time-to-market and driving innovation, new business, and revenue.

HIGH-LEVEL PRODUCT COMPARISON

	 	
Real Time Operating System (RTOS)	QNX Software Development Platform (SDP 8.0); QNX Neutrino (v7)	VxWorks 7
	QNX OS For Safety	VxWorks Cert Edition; VxWorks 653
Hypervisor	QNX Hypervisor	Helix Virtualization Platform
Cloud enablement	SDP, Hypervisor on Graviton EC 2 (QNX Accelerate)	VxWorks on Graviton EC2
DevSecOps	-	Wind River Studio
OCI Containers & K8s Orchestration	-	VxWorks
Automotive-specific products	BlackBerry IVY	Aptiv Edge Intelligence Middleware
	QNX Sound	Aptiv Sound Framework (ASF)
	QNX Platform for ADAS	Aptiv ADAS Gen 6 platform

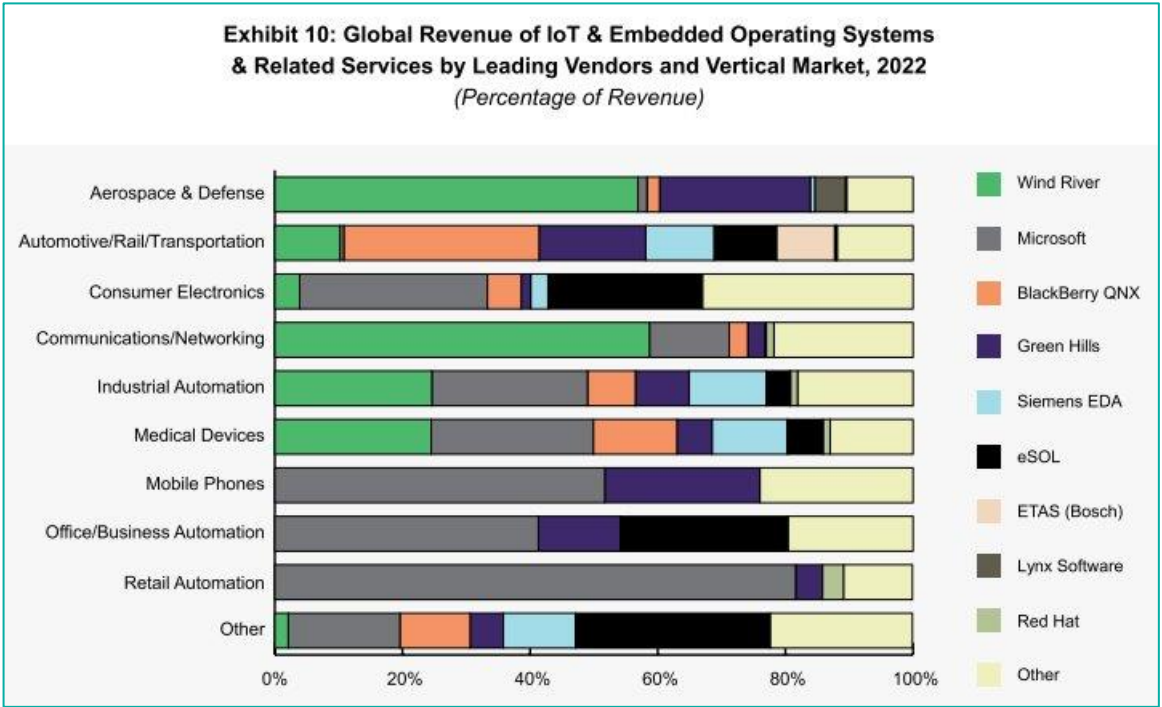
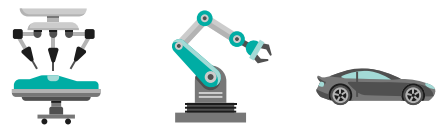
WIND RIVER HOLISTIC APPROACH

WINDRIVER

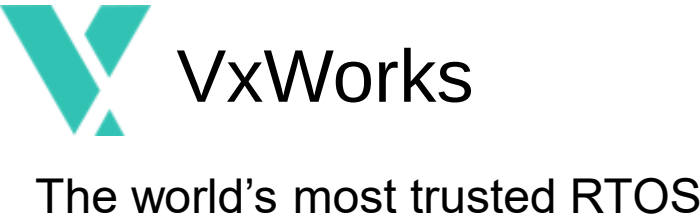


A holistic approach – Wind River Edge, Studio and Aptiv middleware is **foundational**

A VERTICAL VIEW & MARKET POSITION



Source: The Global Market for IoT & Embedded Operating Systems, VDC Research, Oct 2023



CONSIDERATIONS

- Real-Time Operating Systems
- Assessment of Current Systems
- Migration Strategy
 - Incremental vs. Big Bang
- Porting Applications & Drivers
- Test & Validation
- Tooling and Support
- Case Studies and Success Stories
- Q&A Session
- Next Steps

REAL-TIME OPERATING SYSTEM CONSIDERATIONS

Kernel Footprint

Interrupts

Multi-Tasking

Task Context Switching

Virtual Memory Support



COMPARISON BETWEEN QNX AND VXWORKS

Architecture

- QNX's microkernel architecture is inherently different from VxWorks' primarily monolithic approach, impacting system stability and security.

Development Tools

- Both offer powerful, Eclipse-based IDEs tailored to their ecosystems.

Real-Time Performance

- Both are leaders in providing deterministic, real-time performance for critical applications.

Networking and IoT

- While both support connected applications, their approaches and tools for IoT and networking can differ.

Security and Safety

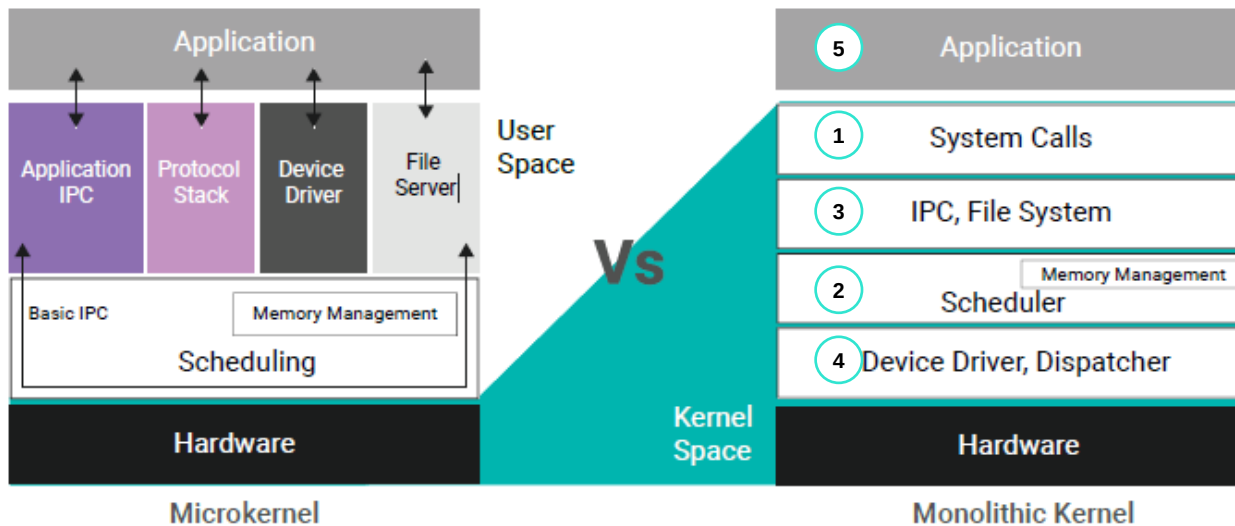
- Both prioritize security and safety, with numerous certifications to back their capabilities, but their specific features and implementations vary.

Market Focus

- Each has its strengths in different sectors, influenced by their historical use cases and core capabilities.

ARCHITECTURAL CONSIDERATIONS NECESSITATE PORTING OF APPLICATIONS AND UNDERLYING CODE

QNX and VxWorks use different architectures...



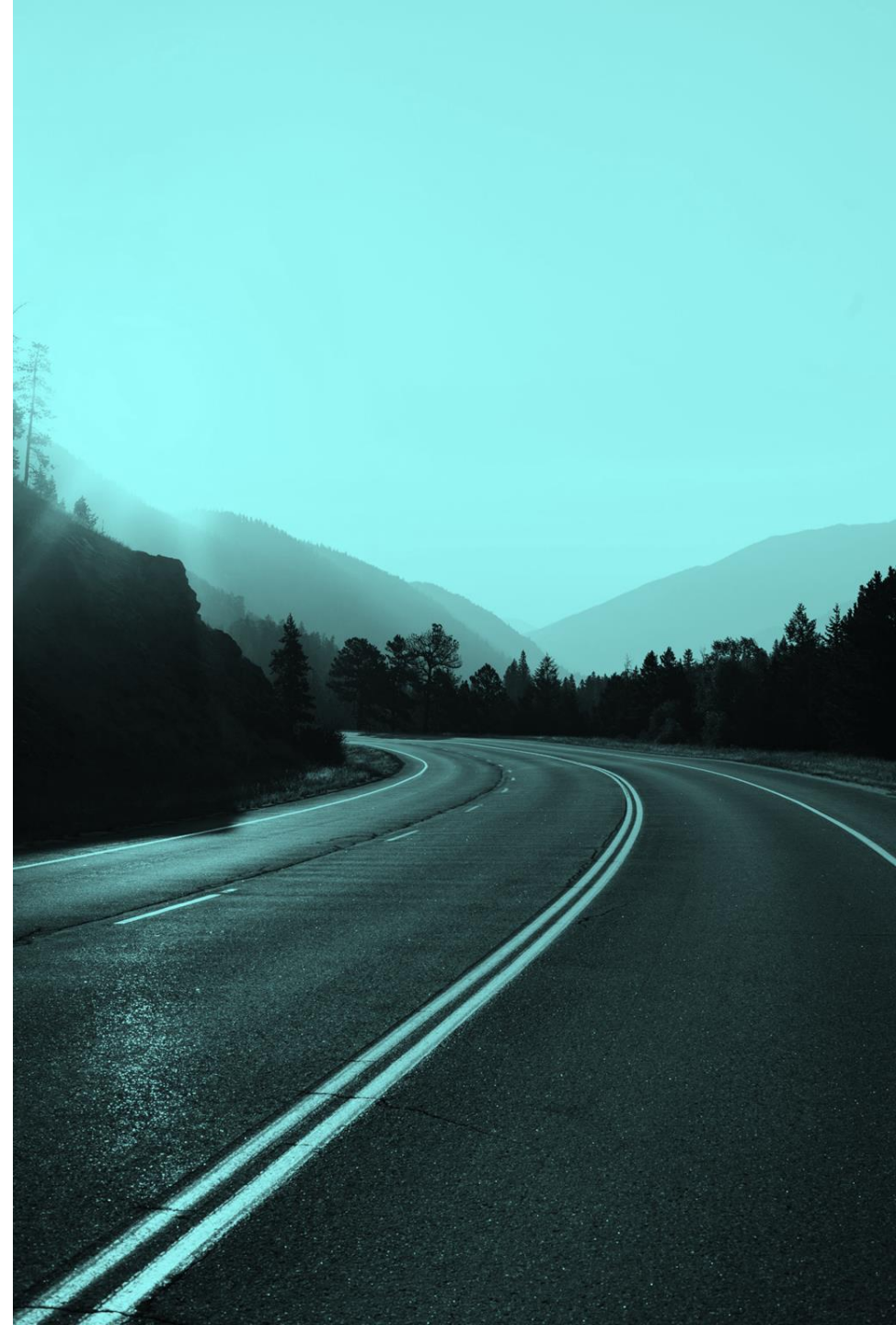
Easy porting and optimized lifecycle management

...necessitating porting of multiple components

- 1 Device drivers**
Will have to be re-written
- 2 System Calls and APIs**
APIs are quite different so must map respective routines
- 3 Multithreading and Synchronization**
From threads to tasks
- 4 Network Programming**
Similar networking structures
- 5 Applications**
Recompile, link and run due to being hardware agnostic

MIGRATING APPLICATIONS

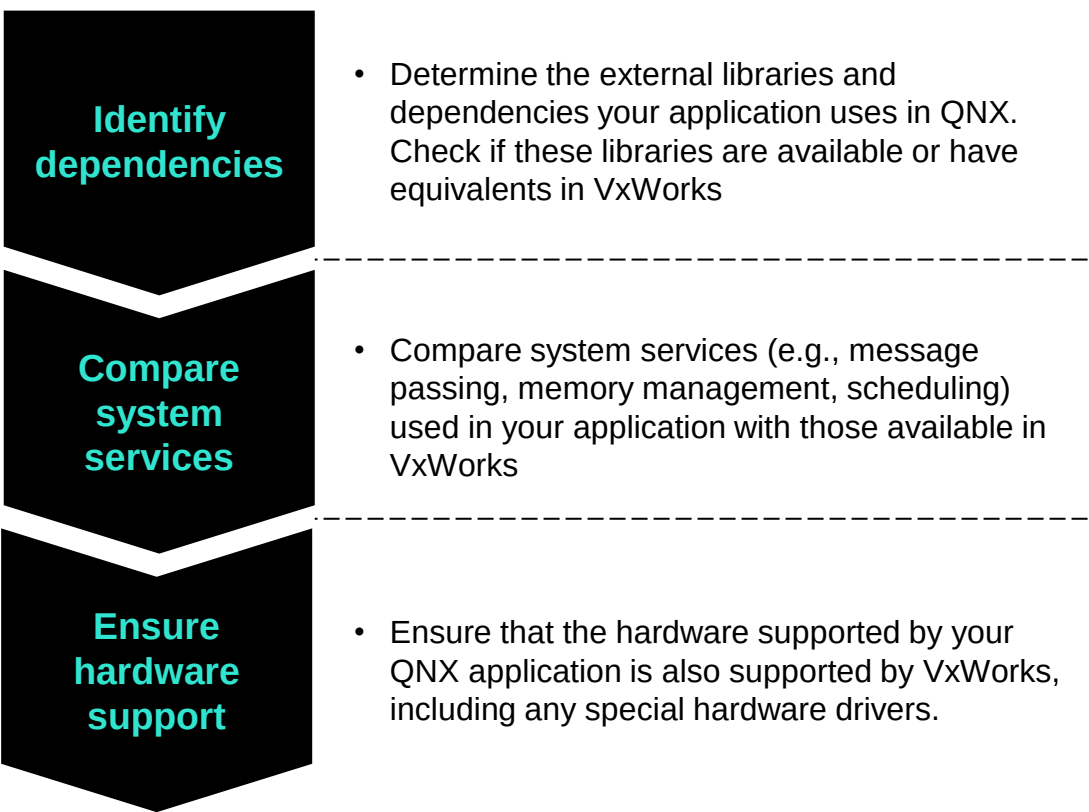
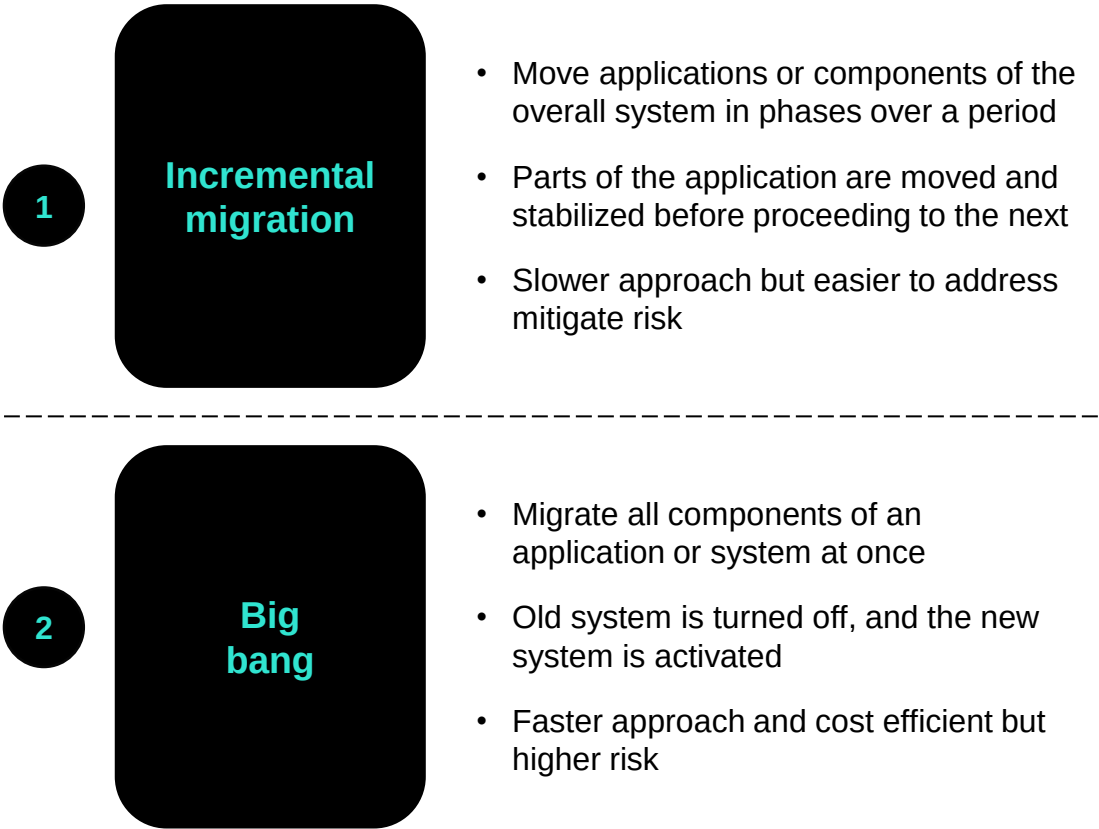
When organizations decide to move their applications from one environment to another — the strategy they choose for migration plays a critical role in the operation's success and efficiency.



TWO APPROACHES TO PORTING WITH MULTIPLE STEPS IN EACH

2 approaches ...

... but same strategies / considerations apply



BENEFITS OF EACH MIGRATION STRATEGY

INCREMENTAL

- Risk Mitigation
- Continuity of Operations
- Feedback and Adjustment
- Resource Management

BIG BANG

- Simplicity
- Quick Transition
- Cost Efficiency



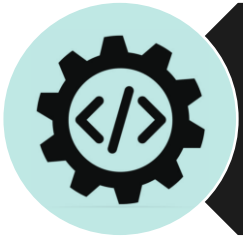
CODE PORTING

Falls into five categories:



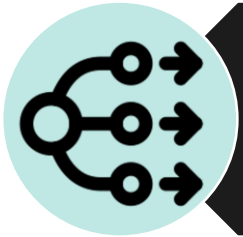
Device Drivers

- Will have to be rewritten



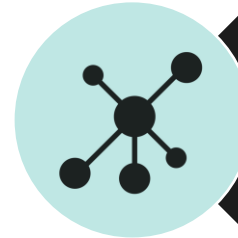
System Calls and APIs

- APIs are quite different so must map respective routines



Multithreading and Synchronization

- From threads to tasks



Network Programming

- Similar networking structures



Application

- Recompile, link and run due to being hardware agnostic

VXWORKS DEBUGGER OVERVIEW



Supports debugging of both Downloadable Kernel Modules (DKMs) and Real-Time Processes (RTP)

Has the capability of debugging interrupt service routines (ISRs)

Can be used to debug both uniprocessor (UP) and symmetric multiprocessor (SMP) systems

Uses Target Communication Framework (TCF) to communicate



DEBUGGING MODES

APPLICATION MODE

Most common for user mode and kernel mode apps

A breakpoint will stop the task that hit it

Not suitable for ISRs

To include application debugging capability in the target, the VIP must have:

INCLUDE_DEBUG_AGENT

To automatically start the debug agent running after boot, include:

INCLUDE_DEBUG_AGENT_START

STOP MODE

Can be used for system or application debugging

A hit on a break point will stop the target

Can be used to debug ISRs

To include stop mode debugging capability in the target, the VIP must have:

INCLUDE_STOP_MODE_AGENT

INCLUDE_STOP_MODE_AGENT_COMM_TYPE

To automatically start the debug agent running after boot, include:

INCLUDE_STOP_MODE_AGENT_START

TOOLING & SUPPORT



Software Downloads

Download your product software and updates



Installation and Licensing

Access installation and licensing documentation



Documentation

Access documentation, tutorials, defects, and patches



Training

Get expert training when and how you want it



Knowledge Forum

Share questions and answers about Wind River products

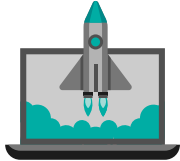
CASE STUDIES & SUCCESS STORIES

Can be found here:

<https://www.windriver.com/success-stories>



BEYOND MIGRATION



Release cadence

- Two VxWorks releases per year
- Upgrades included in pricing
- Provides more choice as to when to upgrade



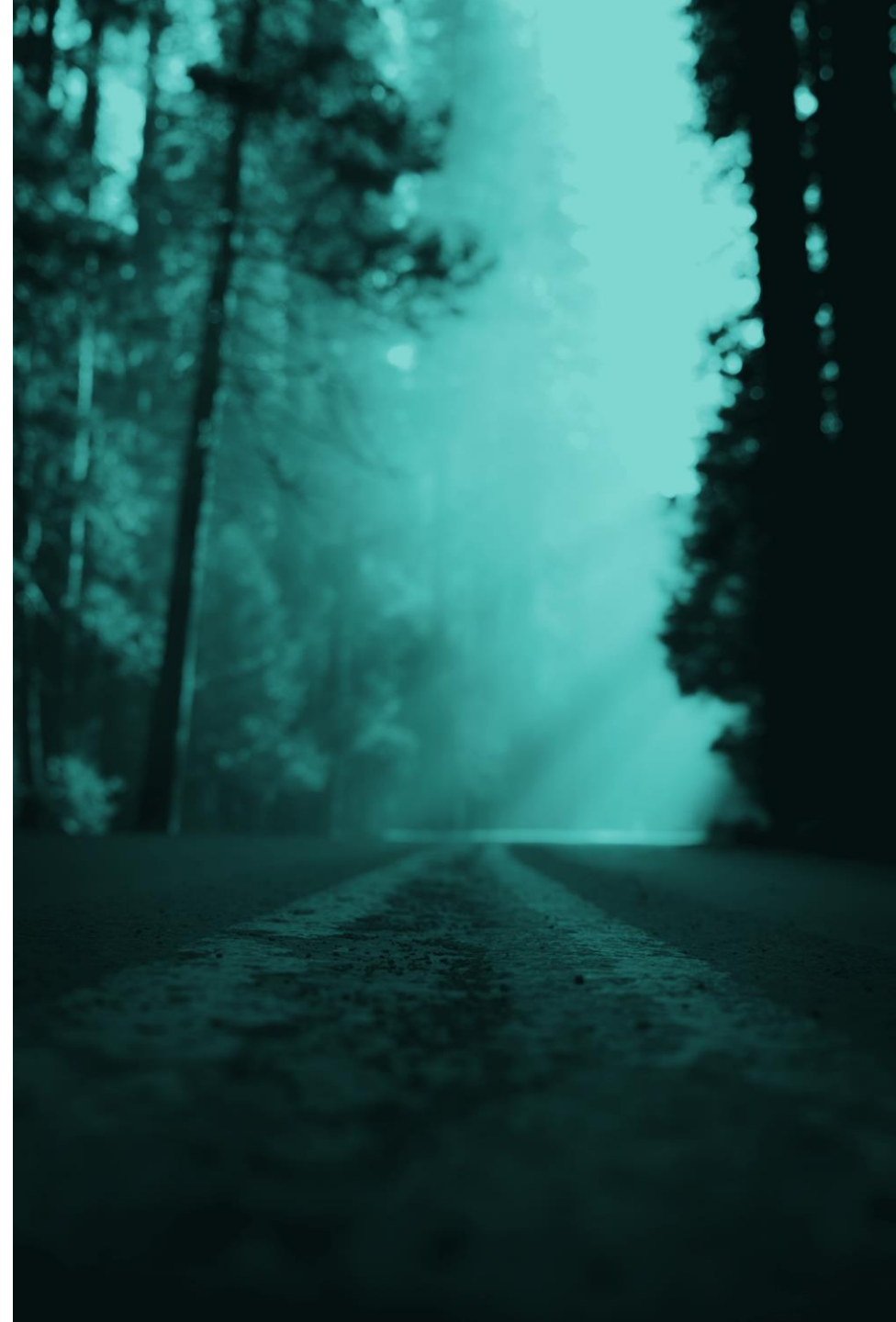
Wider portfolio options – one-stop-shop

- Scale from VxWorks RTOS into mixed criticality with Linux / safety Linux offerings – all on Helix platform
- Adopt ‘shift left’ development & DevSecOps methodologies at development teams’ pace



On-going leadership & innovation

- TSN – natively in VxWorks, supporting the standards you need
- Deployment / orchestration via OCI containers and K8s
- Specialized projects via Professional Services



WIND RIVER PROFESSIONAL SERVICES



Safety Certification	Over 600 safety-certified projects delivered in all industries	DevOps/CI/CD	Enablement and integration of project assets and tools for DevOps and CI/CD
Security & Compliance	Lifecycle security and compliance services from start to finish	Simulation & Modeling	Build digital twins, integrate and optimize tools, and customize models for any hardware system
Real-Time and Deterministic Systems	Services enabling VxWorks and Helix Platform	Customer Success	Dedicated resources to ensure your success using Wind River products across projects
Linux and Open Source	16+ years of open-source leadership including all Linux distributions and Android	Premium Support	Lifecycle support options to meet the unique develop, deploy, and operate requirements of your design
Architectural Design	Design services for the unique development and deployment requirements of embedded systems	Education and Training	Knowledge transfer from our experts to yours through multiple delivery formats
Board Support Packages	Customized BSPs for VxWorks and Linux running on all hardware architectures	Managed Services	Platform monitoring, maintenance, and management services for security, stability, and high availability

WINDRVR

Q&A

